PROGRAM 7: Summarization of an Engineering Document

AIM

To summarize a lengthy engineering document using a pretrained language model.

ALGORITHM

1. Import the required libraries.
2. Load a pretrained summarization model.
3. Accept engineering document text.
4. Generate a summary.
5. Display the summarized content.

CODE

```
from transformers import pipeline


summarizer = pipeline(
    "summarization",
    model="sshleifer/distilbart-cnn-12-6"
)


document = """
Artificial Intelligence is an important technology in engineering.
It allows machines to perform tasks that require human intelligence.

Machine learning is a part of artificial intelligence.
It helps computers learn from data and make predictions.

Deep learning uses neural networks to solve complex problems.
AI is widely used in healthcare, robotics, transportation,
education and industrial automation.

AI helps engineers improve productivity, reduce errors
and develop intelligent systems.
```

```
"""

summary = summarizer(
    document,
    max_length=80,
    min_length=25,
    do_sample=False,
    truncation=True
)

print("Original Document:")
print(document)

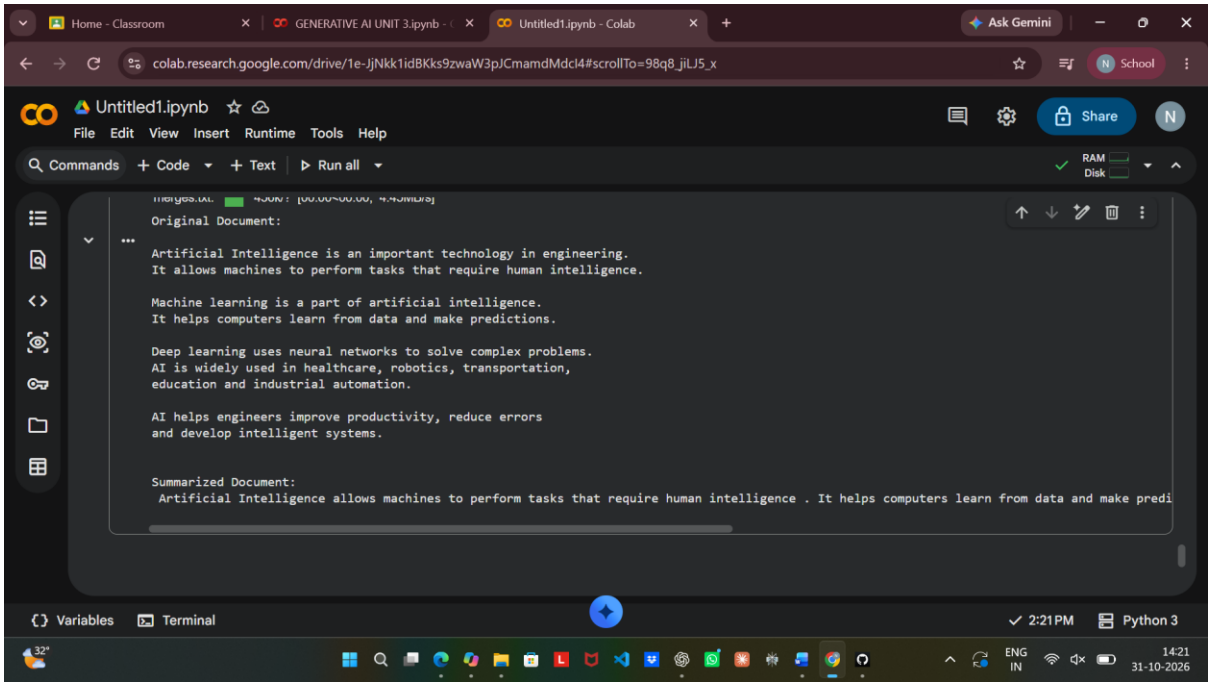
print("\nSummarized Document:")
print(summary[0]["summary_text"])
```

INPUT

An engineering document discussing Artificial Intelligence, Machine Learning and Deep Learning.

OUTPUT

A shorter summary containing the main ideas of the original document is displayed.



RESULT

Thus, an engineering document was successfully summarized using a pretrained language model.